

CSE 220 -- Practice Set: Fourier Series & CFT (10 problems)

General rules for all problems (same as your real online exams):

No `np.fft`, `scipy.fft`, or any built-in transform routine, anywhere.

All integrals via `np.trapezoid` (or `np.trapz` if that's what your numpy has).

Where a problem says "using the OOP framework," build a small class rather than manipulating plain arrays by hand -- that's usually where marks live.

Set a real timer per problem. Getting the math right in your head *before* you start typing is most of the time budget in a 25-35 min slot.

Problem 1 -- FS Time-Shift Property

Time: 30 min

Background: For a periodic complex signal $f(t)$ with FS coefficients c_n , if $g(t) = f(t - t_0)$, its coefficients are $d_n = c_n \cdot e^{(-j n \omega t_0)}$, where $\omega = 2\pi/T$.

Task: Using your `FourierEpicycles` class (or a thin wrapper around it), load any provided shape (or use a synthetic closed curve `z(t) = cos(t) + 0.6j*sin(2t)`, $t \in [0, 2\pi]$) as $f(t)$. Let $t_0 = T/6$. Build $g(t) = f(t - t_0)$ by wrapping the underlying function of t -- do **not** `np.roll` the sampled array. Compute c_n (f) and d_n (g) for $n = -100..100$. Verify $d_n \approx c_n \cdot e^{(-jn\omega t_0)}$ for every n (MSE of magnitude and of unwrapped phase). Plot $|c_n|$ and $|d_n|$ as overlaid stem plots vs n .

Problem 2 -- FS Linearity / Superposition

Time: 25 min

Background: FS is linear: coefficients of $\alpha \cdot f + \beta \cdot g$ are $\alpha \cdot c_n(f) + \beta \cdot c_n(g)$, *provided f and g share the same period T* .

Task: Load two different SVG shapes with the same `num_points` (so they share the same t -grid and period). Let $\alpha = \beta = 0.5$ and build $h(t) = \alpha \cdot z_1(t) + \beta \cdot z_2(t)$. Compute $c_n(h)$ directly, then separately compute $c_n(z_1)$, $c_n(z_2)$, and combine $\alpha \cdot c_n(z_1) + \beta \cdot c_n(z_2)$. Verify equality for $n = -100..100$ (MSE). Bonus: render the epicycle animation of $h(t)$ -- does it look like a blend of the two shapes?

Problem 3 -- FS Differentiation Property

Time: 30 min

Background: If $x(t) = \sum c_n e^{(jn\omega t)}$, then dx/dt has FS coefficients $j n \omega \cdot c_n$.

Task: Use a *smooth* synthetic periodic signal (avoid sharp-cornered shapes here): $x(t) = \cos(t) + 0.3j \cdot \sin(3t)$, $t \in [0, 2\pi]$, 3000 samples. Derive $y(t) = dx/dt$ analytically by hand. Compute $c_n(x)$ and $d_n(y)$ for $n = -50..50$. Verify $d_n \approx j n \omega \cdot c_n$ ($\omega=1$ here). Separately comment: why must d_0 (the $n=0$ coefficient of *any* periodic signal's derivative) always equal zero, regardless of what $x(t)$ is?

Problem 4 -- FS Parseval's Theorem

Time: 20 min

Background: $(1/T) \int_0^T |f(t)|^2 dt = \sum_n |c_n|^2$.

Task: Take any SVG shape's $z(t)$. Compute the LHS numerically via `trapezoid` of $|z(t)|^2$ over one period.

Compute the RHS by summing $|c_n|^2$ for $n = -150..150$. Report both values and the relative error. Then plot the *cumulative* sum of $|c_n|^2$ against the number of harmonics included -- how many harmonics does it actually take to get within 1% of the LHS?

Problem 5 -- FS Conjugate Symmetry for Real Signals

Time: 20 min

Background: If $f(t)$ is real-valued, $c_{-n} = \text{conjugate}(c_n)$ for all n .

Task: Build a real-valued periodic signal, e.g. a square wave $f(t) = \text{sign}(\sin(t))$, $t \in [0, 2\pi]$, 3000 samples (zero imaginary part). Compute its FS coefficients for $n = 1..50$ and verify $c_{-n} = \text{conjugate}(c_n)$ numerically (MSE of the complex difference). Then reconstruct $f_N(t)$ two ways: (a) the full symmetric complex sum over $n = -N..N$, and (b) $c_0 + 2 \cdot \sum_{n=1}^N \text{Re}[c_n \cdot e^{(jn\omega t)}]$ using only $n \geq 0$. Confirm both give the same real-valued result, and explain why in 2-3 sentences.

Problem 6 -- CFT Linearity + Real (Cosine) Modulation

Time: 30 min

Background: CFT is linear. Real modulation splits the spectrum: $F\{x(t) \cdot \cos(2\pi f_0 t)\} = 0.5 \cdot [X(f-f_0) + X(f+f_0)]$ -- different from multiplying by a complex exponential, which just shifts the whole spectrum by f_0 .

Task: Build $x_1(t) = e^{-(t^2)}$ (gaussian) and $x_2(t) = \text{rect pulse}$ (amplitude 1 for $|t| \leq 1$, else 0). $t \in [-8, 8]$, 3000 samples. Let $y(t) = 2 \cdot x_1(t) + x_2(t) \cdot \cos(2\pi \cdot 3 \cdot t)$, $f_0 = 3$. Write your own `compute_cft` (functional style is fine here, OOP not required). Verify $Y(f) \approx 2 \cdot X_1(f) + 0.5 \cdot [X_2(f-f_0) + X_2(f+f_0)]$ over $f \in [-10, 10]$, 1000 points. Report magnitude and phase MSE (mask/unwrap as needed).

Problem 7 -- CFT Convolution Theorem

Time: 35 min

Background: $(x_1 * x_2)(t) = \int x_1(\tau) x_2(t-\tau) d\tau$, and $F\{x_1 * x_2\}(f) = X_1(f) \cdot X_2(f)$.

Task: $x_1(t) = \text{rect pulse}$, width 1, amplitude 1 ($|t| \leq 0.5$). $x_2(t) = \text{triangular pulse}$, width 1, peak 1 at $t=0$ (linear to 0 at $|t|=0.5$). Implement `conv_numeric(t, x1, x2)` that computes $y(t) = (x_1 * x_2)(t)$ at each sample via a nested `trapezoid` integral over τ . **Practical tip:** an $N \times N$ convolution this way gets slow fast -- use a coarser grid (~400-500 points) just for this problem, not the full 2000. Compute $Y(f) = \text{compute_cft}(t, y, \text{freqs})$ from the numerically-convolved $y(t)$, and separately $X_1(f) \cdot X_2(f)$ elementwise. Compare via MSE. Note the runtime trade-off you had to make in your answer.

Problem 8 -- 2D CFT Feature Addition: Band-Pass Filter

Time: 25 min

Task: Your offline's `FrequencyFilter` currently implements `high_pass(real, imag, cutoff)`. Add a **new** method `band_pass(real, imag, low, high)` that keeps only frequency components whose radius from the spectrum center lies in `[low, high]` (same pixel-index units as `high_pass`), zeroing everything outside that band. Apply it with `(low, high) = (5, 20)` to any grayscale image. Reconstruct via `InverseCFT2D` and compare side-by-side with the pure high-pass result (`cutoff=15`). In 2-3 sentences, what visual information does the band-pass keep that neither the low-pass nor high-pass alone shows?

Problem 9 -- 2D CFT Spatial Shift Property

Time: 25 min

Background: Shifting an image by (x_0, y_0) multiplies its 2D-CFT by a linear phase ramp: $F_{\text{shifted}}(u,v) = F(u,v) \cdot e^{-j2\pi(u \cdot x_0 + v \cdot y_0)}$; magnitude is unchanged.

Task: To keep runtime manageable, build a small **synthetic** test image (e.g. 64×64 , zero everywhere except a bright 8×8 patch) rather than a full photo. Build a second image with the same patch shifted by (x_0, y_0) in continuous coordinates. Compute `CFT2D` for both. Verify $|F_{\text{shifted}}| \approx |F|$, and that the phase difference at a handful of sample (u,v) points matches the predicted ramp $-2\pi(u \cdot x_0 + v \cdot y_0)$. Report MSE for magnitude and for the predicted-vs-measured phase difference.

Problem 10 -- Creative: Gibbs Phenomenon & Convergence Study

Time: 30 min

Task: Using `star.svg` (or any sharp-cornered shape) and your `FourierEpicycles` class, compute the reconstruction $f_N(t)$ for $N \in \{5, 20, 50, 100, 150, 300\}$. For each N , compute MSE between $f_N(t)$ and the true sampled $z(t)$. Plot MSE vs N (log-scale y-axis). Then zoom into the neighborhood of one sharp corner and overlay $f_N(t)$ for your two largest N values -- does the overshoot near the corner shrink as N grows, or does it converge to a roughly constant percentage overshoot? Write 3-4 sentences on what you observe and why, and note how you'd expect `circle.svg` (no sharp corners) to behave differently.

Coverage check

Fourier Series: Problems 1-5 (shift, linearity, differentiation, Parseval, conjugate symmetry)

1D CFT: Problems 6-7 (linearity/modulation, convolution)

2D CFT: Problems 8-9 (feature addition, spatial shift)

Conceptual/creative: Problem 10 (Gibbs phenomenon)